2.4 Least Common Multiple (L.C.M)

L. C. M of two or more given numbers is the smallest number which is divisible by all the given numbers.

Multiples

How many steps can a rabbit jump if it is jumping 4 steps for 10 times?

4, 8, 12, 16, 20, 24, 28, 32, 36, 40

If we add 4 to each number we will get 40.

Try this

5, 10, 15, ----, ----, -----, -----, -------

Do you know?

Each number is the first multiple of that number.

There is a relation between the multiples and multiplication table of a number.

Common Multiples:-

The Multiples that are common to two numbers are their common multiples.

Example

Find the common multiples of 9, 12

Multiples of 9 - 9, 18, 27, 36, 45, 54, 63, 72, 81, 90,...

Multiples of 12 - 12, 24, 36, 48, 60, 72, 84, 96, 108, 120,...

The common multiples of 9, 12 are **36, 72,...**

Example

Find L.C.M of 4 and 6

Multiples of 4 - 4, 8, 12, 16, 20, 24, 28, 32, 36,...

Multiples of 6 - 6, 12, 18, 24, 30, 36, 42, 48, 54,...

Common multiples of 4, 6 are 12, 24, 36,...

L.C.M of 4, 6 is 12.

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